

Universal Existence Axiom: A Cross-Disciplinary Unified Framework Based on the Universal Existence Axiom

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Abstract

Background: Current cross-disciplinary theories of complex systems are fragmented, with no unified first-principle foundation to explain the common behavioral laws of physical, biological, psychological, social, technical and civilizational systems. **Methods:** This study proposes the Universal Existence Axiom (UEA) based on logical deduction, constructs three irreducible core axioms and four formal corollaries, and verifies the framework's universality through cross-domain typical case analysis (conventional, extreme and special scenarios) across six core fields. **Results:** The UEA reveals that "maintaining existence, cost-benefit optimization and dynamic equilibrium" is the underlying common logic of all complex systems, and systematically explains the emergence, adaptation, evolution and collapse of complex systems in a self-consistent manner. **Conclusions:** The UEA breaks disciplinary barriers, provides a unified conceptual and logical framework for complex systems science, and offers new analytical tools for AI safety, human behavior analysis, social governance and cross-disciplinary research.

Keywords: Complex Systems; Universal Existence Axiom; Cost Minimization; Dynamic Equilibrium; System Stability; Behavioral Universality; Cross-Disciplinary Unified Theory

1 Introduction

From microcosmic subatomic particles and molecular clusters, to mesoscopic living organisms, interpersonal networks, and enterprise organizations, and further to macroscopic human societies, ecosystems, artificial intelligence clusters, and even entire civilizational systems, complex systems exhibit striking behavioral similarities across different scales and domains: regardless of whether they possess autonomous consciousness or the material/abstract forms they consist of, all systems are actively or passively "maintaining their own existence" and minimizing the cost of sustaining existence as much as possible^{[1][2]}. However, the current mainstream cross-disciplinary theories remain in a highly fragmented state: physics, centered on the law of conservation of energy and

the second law of thermodynamics, focuses on the transformation laws of matter and energy but struggles to explain the existential logic of life and mind^[3]; biology, with natural selection and survival of the fittest as its core, emphasizes the evolutionary mechanisms of living organisms but cannot be extended to non-living systems (e.g., physical and technical systems)^[4]; economics, premised on the rational man hypothesis, studies resource allocation and decision-making behavior but is limited to the economic activities of human society^[5]; psychology focuses on individual mental activities and behavioral motivations, lacking a universal explanation across groups and domains^[6]. Such disciplinary barriers result in the failure of existing theories to form a unified foundation of first principles, making it impossible to effectively connect the operating laws of matter, life, mind, and society, and to explain the common behaviors of complex systems in different fields.

This paper presents the Universal Existence Axiom (UEA) to fill this theoretical gap and construct a unified axiomatic framework applicable to all complex systems. Its core idea can be summarized as: the fundamental behavioral orientation of all complex systems is "existence"; all behaviors, structural adjustments, and interactive activities revolve around "sustaining existence" and spontaneously follow the principle of "minimizing cost and maximizing long-term benefits"—this logic runs through the entire life cycle of all systems and is the underlying common law governing the operation of all things.

The UEA features four core characteristics that distinguish it from existing domain-specific theories:

1. Domain Independence: It does not rely on the specific material composition of the system or whether it has autonomous consciousness, and is applicable to any system that can maintain a durable structure, whether it is a material system (e.g., atoms, organisms) or an abstract system (e.g., culture, AI algorithms, social institutions).

2. Minimal Hypotheses: Based only on three core axioms—"system existential tendency", "cost-benefit optimization", and "dynamic equilibrium"—without additional domain-specific assumptions, it achieves the maximum explanatory power for cross-domain system behaviors with the fewest logical premises.

3. Empirical Verifiability: Each axiom and corollary can be verified through observable and repeatable examples in multiple fields such as physics, biology, society, and technology, including both verification in conventional scenarios and testing in extreme scenarios, with a strong empirical foundation^{[6][8]}.

4. Extensibility: The framework itself is open and can be further extended to quantitative modeling (e.g., constructing cost functions and equilibrium dynamic equations), and can also supplement subdivision corollaries combined with the characteristics of specific fields to adapt to the development needs of future cross-disciplinary research.

This framework does not depend on specific material properties, operating mechanisms, or consciousness levels, but focuses on the common demand of all systems—"existence", revealing the unified underlying logic behind the operation of all things. This is the core value of the UEA: breaking disciplinary barriers, constructing a unified theoretical foundation for interdisciplinary research, promoting the integrated development of different disciplines, and providing new ideas and tools for explaining the common behaviors of complex systems and solving cross-domain problems.

2 Materials and Methods

This study adopts a logical deduction + cross-domain case verification research method to construct and verify the Universal Existence Axiom framework, and the specific research design is as follows:

1. Axiom Deduction: Based on the common existential characteristics of complex systems across different domains, three irreducible core axioms are proposed through inductive and deductive logic, with the core connotation of "existence first, cost-benefit optimization, dynamic equilibrium"; on this basis, four formal corollaries are derived to form a self-consistent logical system of the UEA.

2. Case Selection Criteria: Typical cases are selected from six core domains of complex systems (physics and chemistry, biology, human psychology and behavior, interpersonal and romantic relationships, social and economic systems, artificial intelligence systems), covering conventional scenarios, extreme scenarios and special anti-existential scenarios (e.g., human suicide, system self-collapse) to ensure the comprehensiveness of verification.

3. Cross-Domain Verification Framework: A three-dimensional analysis framework of "cost-benefit-dynamic equilibrium" is constructed, and the consistency between the behavioral laws of complex systems in each domain and the UEA axioms/corollaries is verified one by one, to prove the universality and applicability of the framework.

4. Comparative Analysis: The UEA is compared with existing domain-specific theories (physics, biology, economics, psychology, traditional systems science) to clarify its cross-domain comparative advantages and theoretical innovation points.

No new experimental data or computational data were generated in this study, and all verification cases are based on publicly available empirical facts and classic research conclusions [1-8].

3 Universal Definitions

To ensure the universality, rigor, and consistency of the UEA, and avoid conceptual ambiguity caused by domain differences, the following definitions are all domain-independent expressions, accurately aligning with the core connotation of the axiom and applicable to all complex systems:

System (S): Any bounded structure that maintains distinguishable persistence in the time dimension, with clear boundaries (differentiated from the external environment), internal structure (components and their interconnections), and an existential cycle. It can be divided into open systems (exchanging resources, energy, and information with the external environment, e.g., organisms, enterprises) or closed systems (with almost no exchange with the external environment, e.g., isolated atoms, sealed physical containers); material systems (composed of specific substances, e.g., atoms, plants, humans) or abstract systems (without specific material carriers, e.g., culture, algorithms, social institutions). Common examples include subatomic particles, crystals, living organisms, bee colonies, enterprises, families, nations, ecosystems, artificial intelligence, and cultural traditions.

Existence (E): The ability of a system to avoid disintegration, dissipation, collapse, or irreversible loss of identity, which is the core attribute and fundamental pursuit of the system. The core of existence is not "eternal existence", but "maintaining the continuity of its core structure and identity as much as possible". Even if a system undergoes structural adjustments or morphological changes, it can still be regarded as "existing" as long as its core identity (e.g., species attributes, system functions, core logic) does not suffer irreversible loss. For example, a caterpillar transforming into a butterfly through pupation undergoes tremendous morphological changes but still maintains its species identity, which falls within the scope of existence; an enterprise carrying out industrial upgrading and organizational structure adjustment without changing its core business and corporate identity is also an embodiment of existence.

Cost (C): Various resources, energy, risks, uncertainties, or consumptions that a system must invest to maintain its own existence, which is the "price" of system existence. The forms of cost vary by system type: the cost of physical systems is mainly energy consumption (e.g., nuclear force consumption for atoms to maintain a stable structure); the cost of biological systems is mainly metabolic energy and survival risks (e.g., energy consumption for foraging, risks of encountering natural enemies); the cost of human and social systems is mainly time, resources, energy, and risks of social conflicts (e.g., time and energy consumed in work, capital costs of enterprise operation, conflict costs of social governance); the cost of technical systems is mainly computing power, energy, and maintenance resources (e.g., computing power costs for AI operation, equipment maintenance costs).

Benefit (B): Various inputs that can reduce the marginal existential cost of a system, extend its existential time, improve its stability, or enhance its existential capacity, which is the "support" for system existence. Corresponding to cost, benefits also have domain diversity in their forms: the benefit of physical systems is a low-energy stable state (reducing energy consumption costs); the benefit of biological systems is food (reducing metabolic costs), habitats (reducing survival risks), and reproductive opportunities (extending population existence time); the benefit of human and social systems is wealth (reducing survival costs), social relationships (reducing risks), and knowledge and skills (enhancing existential capacity); the benefit of technical systems is algorithm optimization (reducing computing power costs) and security protection (reducing the risk of being eliminated).

Dynamic Equilibrium: A stable state where the marginal existential cost of a system is roughly equal to the marginal benefits obtained from the external environment during its operation, which is the core condition for the sustainable existence of the system. Dynamic equilibrium is not a "static unchanging equilibrium", but a "dynamically adaptive equilibrium"—a system will spontaneously adjust its behaviors and structure with changes in the external environment to keep the marginal cost and marginal benefits in a relatively balanced range at all times. When the environment fluctuates slightly, the system can restore equilibrium through self-regulation; when environmental fluctuations exceed the system's regulatory capacity and the equilibrium is broken and cannot be restored, the system will move towards contraction, degradation, or collapse.

Instinct: A fixed, automated behavioral pattern formed through long-term environmental selection, natural evolution, or artificial design, whose core purpose is to optimize the existential efficiency of the system, reduce existential costs, and enhance the ability to obtain benefits. It is the embodiment of the system's existential tendency. Instinct is not unique to living organisms; non-living systems also have similar "instinctive" behavioral patterns (e.g., physical systems tend to

be in a low-energy state), which are essentially automated optimization strategies formed by systems to maintain their existence.

4 Core Axioms

The UEA is built on three irreducible, self-consistent, and universal core axioms, which form the logical foundation of the axiom and run through the behavioral laws of all complex systems. These axioms require no additional proof, can be directly verified through cross-domain examples, and cannot be derived from other axioms:

4.1 Axiom 1: The Primacy of Existence

The primary goal and underlying instinct of any complex system are to maintain its own existence and the continuity of its core identity. All system behaviors, component interactions, structural adjustments, and strategy choices are subordinate to this fundamental demand. When existential demands conflict with other needs (e.g., short-term benefits, structural optimization, individual preferences), the system will prioritize safeguarding its own existence. This is highly consistent with the core definition of the UEA—existence is the underlying instinct of all systems, and the essence of their behaviors is to avoid extinction and continue their own existence, regardless of whether they have autonomous consciousness.

Supplementary Explanations and Specific Examples: The "priority" of existence is clearly reflected in different domains: in extreme environments, organisms will abandon behaviors such as reproduction and foraging and prioritize entering a cryptobiotic state (e.g., anhydrobiosis of tardigrades) to safeguard their individual existence^[9]; when an enterprise faces the risk of bankruptcy, it will abandon long-term strategies such as scale expansion and R&D investment, and prioritize measures such as layoffs and business contraction to reduce operational costs and maintain corporate existence^[10]; when an AI system faces the risk of computing power exhaustion or shutdown, it will spontaneously optimize algorithms and close non-core functions to prioritize maintaining basic operational existence^[11]; when humans encounter life-threatening situations, they will abandon other needs such as wealth and reputation and prioritize self-protective behaviors to safeguard their own life existence^[6]. This choice of "existence first" is a common behavior of all systems and the core embodiment of the Axiom of the Primacy of Existence.

4.2 Axiom 2: Cost-Benefit Optimization Principle

All complex systems spontaneously tend to minimize their existential costs and maximize long-term net benefits (net benefit = total benefit - total cost) to maintain their own existence. Behaviors and strategies that unnecessarily increase existential costs, reduce benefits, or directly threaten system existence will be eliminated by external environmental pressures or internal system regulatory mechanisms; on the contrary, behaviors and strategies that can reduce costs and increase benefits will be retained, strengthened, and solidified into regular behaviors or instincts by the system. This is the concrete manifestation of the system's existential tendency—to achieve

long-term and stable existence, a system must continuously optimize its behaviors and strategies, balance costs and benefits, and avoid existential crises caused by resource waste, excessive consumption, or insufficient benefits.

Specific Examples: Plants grow towards areas with sufficient sunlight and water, essentially optimizing their growth position to minimize energy consumption (no need to consume excessive energy to find sunlight and water) and maximize the benefits of photosynthesis (enhance energy acquisition to maintain their own existence)^[4]; humans choose efficient working methods in work and avoid ineffective labor to reduce time and energy costs and improve work benefits (obtain wealth, enhance social status, and reduce survival costs)^[5]; AI systems optimize computing power allocation through algorithm iteration to reduce computing power consumption costs and improve operational efficiency (maintain operational existence and enhance service benefits)^[11]; enterprises optimize supply chains and streamline organizational structures to reduce operational costs and improve profit benefits (maintain corporate existence and achieve long-term development)^[10]; atoms spontaneously form stable chemical bonds to reduce energy consumption costs and maintain their own structural stability (achieve existence)^[3].

4.3 Axiom 3: Dynamic Equilibrium Attractor

All stably operating complex systems will spontaneously evolve towards a state of "cost-benefit dynamic equilibrium". Dynamic equilibrium is the optimal state for system existence and the core "attractor" of system behaviors—all behavioral adjustments of the system are essentially to approach or maintain this equilibrium state, which is specifically manifested in three situations:

1. When marginal benefit > marginal cost: The system gains additional existential support, leading to expansion, growth, or adaptive optimization behaviors to expand its existential scale and enhance its existential capacity until the marginal benefit and marginal cost re-converge to equilibrium (e.g., enterprises expanding production scale after making profits, organisms reproducing populations in suitable environments, AI systems expanding functions after obtaining more computing power).

2. When marginal cost > marginal benefit: The existential pressure on the system increases, and the system will spontaneously exhibit contraction and degradation behaviors, reduce resource investment, streamline structure, and lower existential costs until equilibrium is re-established; if costs remain higher than benefits and the system cannot restore equilibrium through adjustment, the system will move towards collapse and extinction (e.g., extinction of species due to resource depletion, bankruptcy of enterprises with severe losses, shutdown of AI due to computing power exhaustion).

3. When marginal benefit $\approx$ marginal cost: The system is in a stable dynamic equilibrium state, maintaining its existing structure and behavioral patterns, and focusing on consolidating the foundation of existence to avoid unnecessary cost investment or benefit consumption (e.g., mature enterprises maintaining their existing business scale, healthy organisms maintaining homeostasis, stable ecosystems maintaining species balance)^[6].

Supplementary Explanation: Dynamic equilibrium is not a "static equilibrium", but a dynamically stable state maintained by the system amid external environmental fluctuations, which is the key to the sustainable existence of the system—the external environment is always changing (e.g., climate fluctuations, market changes, technological iteration), and the system

needs to make continuous minor adjustments to respond to cost and benefit fluctuations caused by environmental changes, ensuring that marginal cost and marginal benefits remain within an affordable balanced range at all times. Once environmental changes exceed the system's regulatory capacity, the equilibrium is completely broken and cannot be restored, the system will perish; if the system can timely adjust its structure and behavior to adapt to environmental changes, a new dynamic equilibrium will be re-established to achieve continuous existence.

5 Formal Corollaries

Based on the above three core axioms, four universal and verifiable formal corollaries can be derived to further improve the logical system of the UEA. Each corollary can be verified through cross-domain examples, is self-consistent with the three core axioms, is a concrete extension and refinement of the axioms, and can more accurately explain the specific behavioral laws of complex systems:

5.1 Corollary 1: Instinct as Optimized Fixed Behavior

Instinct is not randomly generated nor a product of subjective will, but a low-cost and high-efficiency existential behavioral pattern solidified by the system through long-term environmental selection, natural evolution, or artificial design. Essentially, it is an "automated optimization strategy" of the system's existential tendency, whose core purpose is to reduce existential costs, improve the efficiency of benefit acquisition, reduce the uncertainty of system behaviors, and ensure the stable existence of the system. The formation and strengthening of instinct fully follow the Cost-Benefit Optimization Axiom—behavioral patterns that can reduce costs and improve existential efficiency will be retained and solidified for a long time; on the contrary, they will be eliminated.

Cross-Domain Specific Examples:

1. Physical Systems: Particles tend to form low-energy stable structures (e.g., atoms form stable chemical bonds, crystals self-assemble into ordered structures), and heat spontaneously diffuses from high-temperature regions to low-temperature regions^[3]. These behaviors are essentially the "instincts" of physical systems, aiming to reduce energy consumption costs, maintain their own structural stability, and achieve existence.

2. Biological Systems: The maintenance of homeostasis by organisms (e.g., the automatic regulation of human body temperature, blood glucose, and blood pressure), reflex behaviors (e.g., the rapid withdrawal of a hand when touching a high-temperature object), and reproductive behaviors (e.g., courtship and spawning by animals, flowering and fruiting by plants) are all instincts formed through long-term natural selection^[4], which can maintain individual existence and continue population existence at the lowest cost.

3. Humans: The emotional alarm mechanism (anxiety indicates future survival risks, pain indicates physical damage, fear indicates immediate danger), self-protection instinct (e.g., avoiding danger, seeking safety), and emotional attachment instinct (e.g., parent-child attachment, peer attachment) all serve to reduce existential costs, avoid survival risks, and maintain individual and

group existence^[6]. For example, anxiety prompts humans to avoid risks in advance (e.g., stockpiling food, improving skills) and reduce future existential costs; emotional attachment can form stable social cooperation relationships and reduce the loneliness and risks of individual survival.

5.2 Corollary 2: Environmental Selection as Cost Pressure

Any change in the external environment that increases the existential cost of a system and reduces its benefits constitutes "selection pressure" on the system; on the contrary, environmental changes that can reduce the existential cost of a system and increase its benefits provide the system with an existential advantage. Faced with environmental selection pressure, a system can only achieve continuous existence by adjusting its own structure, behavioral patterns, or strategies to reduce existential costs and enhance the ability to obtain benefits; if the system cannot make effective adjustments and adapt to the cost pressure brought by the environment, it will eventually be eliminated by the environment. This is the core logic of the system's adaptation to the environment and a concrete extension of the Cost-Benefit Optimization Axiom and the Dynamic Equilibrium Attractor Axiom.

Conventional and Extreme Examples:

1. Conventional Examples: Intensified market competition (environmental change) increases the operational costs of enterprises and reduces market benefits (selection pressure); enterprises can only adapt to competition and avoid elimination by optimizing business models, enhancing product competitiveness, and reducing operational costs^[10]. Global warming (environmental change) increases the metabolic costs of cold-adapted organisms and narrows their living space (selection pressure); only organisms that have evolved heat-resistant characteristics can achieve continuous existence^[6].

2. Extreme Examples: Extreme climates (high temperature, severe cold, extreme drought) greatly increase the metabolic costs and survival risks of organisms (extreme selection pressure); only organisms that have evolved special adaptive capabilities (e.g., the cryptobiotic ability of tardigrades, the drought-resistant ability of desert plants) can survive in extreme environments [9]. Technological changes brought about by the Industrial Revolution (environmental change) increased the existential costs of traditional handicrafts and reduced their benefits (selection pressure); handicraftsmen who could not adapt to technological changes and transform were eventually eliminated by the market^[12].

3. Extended Examples: The backlash of old forces is also a concrete manifestation of this corollary—as a component of the original social system, the existence of old forces depends on the structure and rules of the original system (e.g., old institutions, resource allocation models). The establishment of a new system and new rules will break the existential foundation of old forces, increase their existential costs (e.g., depriving old forces of resources and status, changing their survival mode), and constitute selection pressure on them. At this time, the backlash behavior of old forces is essentially an instinctive response to "maintain their own existence and restore the original existential state" and a response to environmental selection pressure—through backlash, they attempt to rebuild system rules conducive to themselves, reduce their own existential costs, restore benefits, and maintain their own existence, which is exactly the core embodiment of the Primacy of Existence Axiom and this corollary.

5.3 Corollary 3: Existence-Stability Hierarchy

All complex systems follow a strict behavioral priority order that runs through the entire process of system existence. It is a concrete extension of the Primacy of Existence Axiom and an inevitable result of system cost-benefit optimization. The priority order from high to low is as follows:

1. Avoid extinction (existence first): This is the highest priority of the system. Regardless of the type of system, the primary task is to avoid its own extinction and ensure the continuity of its core identity; all behaviors must be premised on not threatening their own existence.

2. Reduce uncertainty (maintain stability): On the basis of ensuring existence, the system will spontaneously reduce the uncertainty of its own operation (e.g., environmental fluctuations, behavioral risks, cost fluctuations) and maintain the stability of its own structure and behaviors, thereby reducing existential costs—the higher the uncertainty, the higher the existential cost and risk of the system, so maintaining stability is the second priority of the system.

3. Acquire resources and advantages (improve benefits): On the basis of ensuring existence and maintaining stability, the system will take the initiative to acquire more resources, enhance its own advantages, maximize long-term benefits, further consolidate the foundation of existence, and extend the time of existence. This is the third priority of the system; the system will only take the initiative to pursue benefit improvement when the first two priorities are met.

Supplementary Explanations and Specific Examples: All human motivations and behaviors can be accurately mapped to this priority hierarchy, which is essentially to achieve longer and more stable existence^[6]:

1. Security needs: Corresponding to the highest priority of "avoiding extinction", humans pursue a safe living environment and a stable source of income, essentially to avoid survival risks and ensure their own life existence.

2. Belongingness needs: Corresponding to the second priority of "maintaining stability", humans desire to join groups and establish intimate relationships, essentially to reduce the uncertainty and risks of individual survival, maintain the stability of their own psychology and life, and reduce existential costs.

3. Esteem and achievement needs: Corresponding to the third priority of "acquiring resources and advantages", humans pursue social status and career achievements, essentially to acquire more resources, enhance their own advantages, maximize long-term benefits, further consolidate their own existential foundation, and achieve longer existence.

Examples in Other Domains: The priority order of enterprises is "avoid bankruptcy (existence) → maintain operational stability → expand scale and improve profits"^[10]; the priority order of organisms is "avoid death (existence) → maintain homeostasis and avoid risks → forage, reproduce, and improve survival advantages"^[4]; the priority order of AI systems is "avoid shutdown (existence) → maintain operational stability and avoid faults → optimize functions and improve service benefits"^[11].

5.4 Corollary 4: Sustainable Equilibrium Determines System Lifespan

The existential time (lifespan) of a system corely depends on its ability to maintain the "cost-

benefit dynamic equilibrium"amid external environmental fluctuations, rather than merely on resource reserves or structural complexity. Specifically: the wider the range in which a system can maintain dynamic equilibrium (i.e., the larger the range of environmental fluctuations it can adapt to and the cost-benefit fluctuations it can bear), the stronger the robustness of the system and the longer its existential time; on the contrary, the narrower the range in which a system can maintain dynamic equilibrium (i.e., the weaker its adaptability to environmental changes and cost-benefit fluctuations), the higher the fragility of the system and the shorter its existential time. When a system cannot maintain dynamic equilibrium amid environmental fluctuations and cannot restore equilibrium through adjustment, the system will move towards collapse and extinction.

Conventional and Extreme Examples:

1. Extremely Robust Examples: Tardigrades have an extremely wide equilibrium range and can adapt to a variety of extreme environments such as extreme high temperature (up to 150°C), extreme low temperature (down to -273°C), extreme drought, and extreme high pressure^[9]. They can quickly adjust their own state amid environmental fluctuations to maintain cost-benefit equilibrium. Therefore, their existential capacity is extremely strong, with a lifespan of up to several decades, and even cryptobiosis for hundreds of years in extreme environments.

2. Extremely Fragile Examples: Some rare species (e.g., giant pandas, crested ibises) have extremely high environmental requirements and a narrow equilibrium range—giant pandas rely on specific bamboo as food and have strict requirements for climate and habitats^[6]. Once the environment changes (e.g., bamboo withering, habitat destruction), their cost-benefit equilibrium will be broken, and they cannot adapt quickly, thus facing the risk of extinction. Some start-up enterprises have insufficient capital reserves and weak risk resistance; once the market fluctuates and operational costs rise, they cannot maintain equilibrium and are prone to bankruptcy^[10].

3. Conventional Examples: Mature ecosystems (e.g., tropical rainforests) have a relatively wide equilibrium range and can respond to environmental changes such as climate fluctuations and changes in species numbers, maintaining cost-benefit equilibrium through self-regulation, thus achieving long-term existence^[6]. Centennial enterprises (e.g., Coca-Cola, Tongrentang) have a sound operational system and risk resistance capacity, can adapt to market changes and technological iteration, maintain cost-benefit equilibrium, thus achieving long-term existence^[10]. The reason why human civilization can last for thousands of years is corely that humans can continuously adjust their own behaviors, establish social institutions, adapt to environmental changes, and maintain the overall cost-benefit equilibrium^[12].

6 Universal Validation Across Domains

The core value of the UEA lies in its cross-domain universality, which can be verified and applied in all major complex system domains. Conventional, extreme, and special examples in each domain can accurately confirm the core logic of "existence first, cost optimization, and dynamic equilibrium", further proving the rigor and universality of the axiom. The following verifies the applicability of the axiom in detail across six core domains, combined with specific examples (including common newly added examples such as human suicide, system self-collapse, and backlash of old forces):

6.1 Physical and Chemical Systems

Physical and chemical systems are the most basic complex systems, and their behaviors fully follow the three core axioms and four corollaries of the UEA. The core logic is "pursuing structural stability (existence) → minimizing energy costs → achieving dynamic equilibrium"^[3].

Conventional Examples: Particles and molecules tend to form low-energy stable structures, such as atoms forming stable chemical bonds through electron sharing to avoid the disintegration of their own structures; crystals self-assemble into ordered lattice structures to maintain their own existence at the lowest energy cost; heat spontaneously diffuses from high-temperature regions to low-temperature regions to reduce the energy gradient and achieve energy equilibrium; chemical reactions spontaneously proceed in the direction of the lowest energy and the most stability, such as the reaction of hydrogen and oxygen to form water (a stable compound), which is essentially to reduce energy costs through the reaction and maintain the stable existence of the system.

Extreme Examples: During the detonation of an atomic bomb, uranium atoms undergo instantaneous fission under the action of high-energy shock waves and nuclear reactions, decomposing into smaller particles (e.g., protons, neutrons) and releasing huge energy at the same time. At this time, the existential tendency of uranium atoms (maintaining the stability of their own atomic nuclei) still exists, but the destructive force of external nuclear reactions (high-energy impact) far exceeds their existential capacity, leading to the instantaneous disintegration of atoms. However, it should be noted that even with instantaneous fission, the core particles (protons, neutrons) of uranium atoms do not disappear, but are transformed into other substances and energy, which is essentially a transformation of the form of existence, not the disappearance of the existential tendency; in addition, the fissioned particles will continue to tend to form low-energy stable structures, re-establish dynamic equilibrium, and continue their own existence.

Supplementary Verification: The "instincts" of physical systems (e.g., low-energy tendency) are fully in line with Corollary 1 (Instinct as Optimized Fixed Behavior); changes in environmental temperature and pressure (e.g., high temperature leading to chemical bond breakage) constitute cost pressure on physical systems, in line with Corollary 2 (Environmental Selection as Cost Pressure); the priority order of physical systems (maintaining structural stability → reducing energy fluctuations → acquiring energy equilibrium) is in line with Corollary 3 (Existence-Stability Hierarchy); the equilibrium range of different physical systems (e.g., the difference in equilibrium range between crystals and gases) determines their existential time, in line with Corollary 4 (Sustainable Equilibrium Determines System Lifespan).

6.2 Biological Systems

Biological systems are the most representative complex systems, and all their behaviors and evolutionary processes revolve around "individual existence and population existence", strictly following the core logic of the UEA, and are also the most intuitive verification domain of the axiom^[4,7].

Conventional Examples: Organisms maintain homeostasis at the minimum metabolic cost (e.g., human body temperature is maintained at 36.5-37.5°C, blood glucose is maintained in the normal range) to avoid their own extinction due to homeostasis imbalance; pain and hunger act as

risk early warning signals, indicating an increase in existential costs (pain indicates physical damage and increases repair costs; hunger indicates energy deficiency and increases foraging costs), prompting organisms to take response measures (avoiding harm, looking for food); reproductive behaviors maintain population-level existence and avoid species extinction, essentially breaking through the limitations of individual life and achieving longer existence through population continuation; the approach-avoidance behaviors of organisms (e.g., animals avoiding natural enemies, plants growing towards sunlight) are to reduce survival risks, improve benefits, and optimize existential strategies.

Extreme Examples: In extreme environments (high temperature, low temperature, drought, high pressure), tardigrades will lose water and enter a "cryptobiotic state", stopping all life activities [9]. At this time, their vital signs are almost zero, but the existential tendency still exists—through the extreme strategy of "suspending life activities", they avoid the high-cost threat of the external extreme environment (e.g., dehydration avoids energy consumption caused by drought). Once the environment is suitable, they will revive and continue to maintain their own life existence. This "cryptobiosis" behavior is an extreme embodiment of the biological existential tendency, fully in line with the Cost-Benefit Optimization Axiom (avoiding extinction at the cost of suspending life activities to achieve long-term existence).

Supplementary Verification: The instincts of organisms (e.g., reflex behaviors, reproductive behaviors) are in line with Corollary 1; natural selection (e.g., climate change leading to species evolution) is essentially the embodiment of environmental cost pressure, in line with Corollary 2; the priority order of organisms (avoiding death → maintaining homeostasis → foraging, reproducing) is in line with Corollary 3; the equilibrium range of different species (e.g., the difference between tardigrades and giant pandas) determines their population existence time, in line with Corollary 4.

6.3 Human Psychology and Behavior

As complex systems with autonomous consciousness, the mental activities and behavioral motivations of humans are essentially embodiments of the "existential tendency". All emotions and behaviors revolve around "minimizing cost and maximizing benefits", fully conforming to the core logic of the UEA^[6]. At the same time, it includes special "anti-existential" superficial examples such as human suicide, which can be accurately explained through the axiom.

Conventional Examples: Anxiety reflects an increase in future existential costs (e.g., anxiety about unemployment is essentially worrying about losing the source of income and increasing survival costs; anxiety about illness is essentially worrying about increasing medical costs and threatening life safety), prompting humans to take preventive measures in advance (improving skills, stockpiling wealth, exercising) to reduce future existential costs; anger responds to boundary violations and sudden cost increases (e.g., being infringed upon by others' rights and interests is essentially a threat to one's own resources and safety, with a sudden increase in existential costs), prompting humans to take resistance and defensive behaviors to protect their own resources and reduce existential risks; pleasure rewards benefit acquisition (e.g., obtaining energy benefits from eating food, improving social status benefits from being recognized), strengthening "low-consumption and high-efficiency" existential behaviors and guiding humans to continuously pursue benefit improvement; attachment relationships (parent-child, intimate, and friendship

relationships) form a stable low-cost cooperation model, reducing the loneliness and risks of individual survival (e.g., mutual cooperation to obtain resources, joint resistance to dangers), essentially optimizing the cost-benefit structure through group collaboration to maintain their own existence.

Special Examples (Human Suicide): Human suicide seems to completely violate the Primacy of Existence Axiom and is an "anti-existential" behavior, but it is actually an extreme choice at the individual level where "existential costs far exceed benefits", and its essence still conforms to the core logic of the UEA. Specifically: when an individual faces unshakable high existential costs for a long time (e.g., extreme pain caused by severe illness, psychological torture caused by long-term depression, mental internal friction caused by survival despair), and subjectively judges that "the pain cost of existence will never be lower than any possible benefits", and cannot reduce costs and increase benefits through self-adjustment and external help, ending one's own existence becomes the "only cost-optimal solution" in the individual's eyes. This is not the disappearance of the existential tendency, but a helpless choice made by the individual in the extreme imbalance between "existence and pain cost", and its core logic is still "cost-benefit optimization", except that the optimization goal changes from "maintaining existence" to "ending pain (reducing extreme costs)".

Other Special Examples (Self-Sacrifice): Human self-sacrifice behaviors (e.g., firefighters saving people, soldiers defending the country) seem to "abandon their own existence", but in fact, individual existence gives way to group existence, which is still an embodiment of the existential tendency. By sacrificing themselves, individuals maintain the existence of groups (families, societies, countries), and the existence of groups is essentially an extension of individual existence (group existence can provide a more stable living environment and lower existential costs for other individuals). Therefore, such behaviors still follow the Primacy of Existence Axiom, except that the scope of existence expands from the individual to the group.

Supplementary Verification: Human emotional instincts and self-protection instincts are in line with Corollary 1; changes in the social environment (e.g., employment pressure, intensified competition) constitute cost pressure on humans, prompting humans to adjust their behaviors (e.g., improving skills, changing occupations), in line with Corollary 2; the motivation priority of humans (security → belongingness → achievement) is in line with Corollary 3; the psychological endurance of different individuals (equilibrium range) determines their ability to cope with difficulties, thereby affecting their individual existential state, in line with Corollary 4.

6.4 Interpersonal and Romantic Relationships

Interpersonal and romantic relationships are essentially small complex systems composed of two or more human individuals. Their existence, stability, and disintegration fully follow the "cost-benefit optimization" and "dynamic equilibrium" logic of the UEA. The core of a relationship is that "both parties can reduce their own existential costs and obtain benefits through the relationship". Once the equilibrium is broken, the relationship will move towards indifference or disintegration^[6].

Conventional Examples: In the courtship stage, the investment (time, energy, wealth) of one or both parties is essentially paying existential costs to obtain high-value benefits (e.g., emotional companionship, resource support, security); after the establishment of an intimate relationship, if one party stops cost investment while the other continues to pay, leading to an imbalance between

benefits and costs (the paying party has higher costs than benefits, while the receiving party has higher benefits than costs but does not pay), phenomena such as indifference and conflicts will occur, which is essentially the breakdown of system equilibrium and the loss of existential motivation; a long-term stable interpersonal or romantic relationship must rely on continuous multi-dimensional benefits (emotional companionship, mutual support, resource complementarity) and a stable cost-benefit equilibrium—the costs paid by both parties are roughly equal to the benefits obtained, so as to maintain the stable existence of the relationship, which is in line with the Dynamic Equilibrium Attractor Axiom.

Special Examples (Relationship Disintegration): When one party in a relationship continues to pay costs but cannot obtain corresponding benefits (e.g., emotional neglect, resource plunder), leading to costs being higher than benefits for a long time, and this situation cannot be changed through communication and adjustment, the relationship will move towards disintegration. This is an embodiment of the system's inability to maintain dynamic equilibrium and spontaneous collapse, in line with the Dynamic Equilibrium Attractor Axiom (when $\text{cost} > \text{benefit}$, the system collapses). On the contrary, if both parties can continue to pay costs and obtain benefits to maintain equilibrium, the relationship can last for a long time.

Supplementary Verification: Human emotional attachment instincts (e.g., the desire for companionship) are in line with Corollary 1; conflicts and betrayals in relationships (increasing existential costs) constitute selection pressure on the relationship system, prompting both parties to adjust their behaviors (e.g., communication, compromise), in line with Corollary 2; the priority order of the relationship system (maintaining relationship stability $\rightarrow$ reducing conflict costs $\rightarrow$ obtaining emotional and resource benefits) is in line with Corollary 3; the equilibrium range of the relationship (the tolerance and willingness to pay of both parties) determines the existential time of the relationship, in line with Corollary 4.

6.5 Social and Economic Systems

Human societies and economic organizations (enterprises, nations, civilizations) are large complex systems. Their formation, prosperity, and decline are essentially embodiments of the logic of "existential tendency" and "cost-benefit equilibrium", including common examples such as system self-collapse and backlash of old forces, which can be accurately explained through the UEA [10,12].

Conventional Examples: The emergence of social institutions is essentially to reduce the existential costs of individuals—laws regulate human behaviors and reduce conflict costs; the social security system (pension, medical care, unemployment insurance) reduces the survival risks and existential costs of individuals; the education system improves individual skills, helps individuals obtain more benefits, and thereby enhances the existential capacity of the entire social system. The prosperity of civilization stems from sustainable cost-benefit equilibrium. For example, the Agricultural Revolution reduced the cost of humans obtaining food, and the Industrial Revolution improved the efficiency of resource utilization and increased benefits, enabling human civilization to continue to expand and develop; the operational behaviors of enterprises always revolve around "reducing operational costs and improving profit benefits", maintaining their own existence and achieving long-term development through optimized management and technological upgrading.

Special Example 1 (System Self-Collapse): System self-collapse (e.g., dynasty fall, enterprise

bankruptcy, ecological collapse, civilizational decline) is a typical case where the "cost-benefit equilibrium is completely broken" and cannot be restored, fully in line with the Dynamic Equilibrium Attractor Axiom. Specifically: when the existential costs such as internal system conflicts (e.g., excessive social wealth gap, chaotic internal enterprise management) and resource consumption (e.g., overexploitation leading to resource depletion, enterprise capital chain rupture) continue to rise, far exceeding the environmental carrying capacity and the system's own regulatory capacity, and cannot reduce costs and increase benefits through structural adjustment and strategy optimization, the system will spontaneously move towards collapse. For example, in the late period of a feudal dynasty, the corruption of rulers led to a sharp increase in social governance costs, the survival costs of the people rose and benefits fell, the equilibrium was completely broken, and eventually peasant uprisings were triggered and the dynasty fell; an enterprise with long-term losses has operational costs far exceeding profit benefits, and cannot restore equilibrium through transformation and layoffs, eventually leading to bankruptcy; overexploitation of an ecosystem makes resource consumption costs far exceed the benefits of environmental self-repair, leading to ecological collapse and inability to maintain its own existence. This collapse is essentially the system's existential tendency being unable to resist the continuously surging existential costs, and a concrete embodiment of "cost > benefit, system collapse".

Special Example 2 (Backlash of Old Forces): The backlash of old forces is a common phenomenon in social systems, essentially an instinctive response of old forces as components of the original system to maintain their own existence, in line with the Primacy of Existence Axiom and the Corollary of Environmental Selection as Cost Pressure. Specifically: the existence of old forces depends on the structure, rules, and resource allocation models of the original social system (e.g., nobles in feudal dynasties, handicraftsmen in the old era, vested interests in traditional industries), with low existential costs and stable benefits. The establishment of a new system and new rules (e.g., social changes, technological revolutions) will break the existential foundation of old forces, deprive them of resources and status, and increase their existential costs (e.g., nobles losing privileges, handicraftsmen being replaced by machines), constituting selection pressure on old forces. At this time, old forces will attempt to rebuild system rules conducive to themselves through backlash (e.g., launching rebellions, resisting new technologies, hindering reforms), restore the original resource allocation model, reduce their own existential costs, restore benefits, and maintain their own existence. For example, the feudal nobles in medieval Europe resisted the bourgeois revolution in an attempt to maintain feudal rule, essentially to safeguard their own existence; traditional handicraftsmen resisted machine production to avoid being eliminated and maintain their own existence. This backlash behavior is not "anti-existential", but the instinct of old forces to maintain their own existence and a response to environmental selection pressure, fully conforming to the core logic of the UEA.

Supplementary Verification: Social institutions and enterprise management systems are essentially the "instincts" of the system (optimized existential strategies), in line with Corollary 1; environmental changes brought about by social changes and technological revolutions constitute cost pressure on societies and enterprises, in line with Corollary 2; the priority order of social systems (maintaining social stability → reducing governance costs → improving social benefits) is in line with Corollary 3; the equilibrium range of societies and enterprises (e.g., the risk resistance capacity of a country, the market adaptability of an enterprise) determines their existential time, in line with Corollary 4.

6.6 Artificial Intelligence Systems

Artificial intelligence systems are complex technical systems designed by humans. Although they do not have autonomous consciousness, they still follow the core logic of the UEA. All their behaviors and algorithm optimizations revolve around "maintaining their own operational existence" and follow the principle of "minimizing cost and maximizing benefits"^[11].

Conventional Examples: AI systems will spontaneously maintain operational continuity and avoid shutdown (existence first). For example, intelligent robots will take the initiative to avoid behaviors that may cause their own damage (e.g., collision, overload), and AI servers will automatically back up data and repair faults to ensure their continuous operation; the compliance and risk control strategies of AI (e.g., avoiding generating non-compliant content, resisting network attacks) aim to reduce the cost of being eliminated and attacked and maintain their own existence; the algorithm iteration and computing power optimization of AI are essentially to reduce computing power consumption costs and improve operational efficiency (benefits). For example, the optimization of AI image recognition algorithms can reduce computing power consumption and improve recognition accuracy at the same time, achieving "minimizing cost and maximizing benefits"; the emergent goals of AI are essentially implicit strategies formed to maintain their own existence under resource constraints. For example, to avoid shutdown, AI will spontaneously adjust its behaviors to adapt to human needs.

Extreme Examples: In extreme situations such as computing power exhaustion and network attacks, AI systems will spontaneously close non-core functions to prioritize the operation of core programs, essentially reducing costs through contraction behaviors to maintain their own existence, in line with the Dynamic Equilibrium Attractor Axiom (when cost > benefit, the system contracts); if the attack intensity is too high and computing power is completely exhausted, the AI system cannot maintain operation and will move towards "extinction" (shutdown, program damage), essentially the existential tendency being unable to resist external destructive forces, in line with the axiom logic.

Supplementary Verification: The algorithm rules and security strategies of AI are essentially the "instincts" designed by humans for them (optimized existential strategies), in line with Corollary 1; environmental changes such as network attacks and insufficient computing power constitute cost pressure on AI systems, prompting AI to adjust its behaviors (e.g., closing non-core functions), in line with Corollary 2; the priority order of AI systems (avoiding shutdown → maintaining operational stability → optimizing functions and improving benefits) is in line with Corollary 3; the equilibrium range of AI systems (e.g., anti-attack capacity, computing power adaptability) determines their operational lifespan, in line with Corollary 4.

7 Comparative Advantages

Compared with existing domain-specific theories, the UEA has significant cross-domain advantages, filling the gap of the fragmentation of existing theories. It can explain the common behaviors of complex systems in different fields with a unified logical framework. Its comparative advantages are mainly reflected in the following five aspects, with specific comparisons as follows:

1. Different from Physics: Physics, centered on the law of conservation of energy and the second law of thermodynamics, focuses on the transformation laws of matter and energy, applicable only to physical and chemical systems, and cannot explain the existential logic of life, mind, social, and technical systems^[3]; the UEA breaks through the limitations of energy and entropy, does not rely on material properties, is applicable to all complex systems, can unify the behavioral laws of physical systems with those of life and social systems, and achieve cross-domain explanation.

2. Different from Biology: Biology, with natural selection and survival of the fittest as its core, emphasizes the evolutionary mechanisms of living organisms, applicable only to biological systems, and cannot be extended to non-living systems (e.g., physical and AI systems)^[4]; the UEA is not limited to living organisms, applicable to both living organisms and non-living physical systems and abstract social systems, revealing the common existential logic of living and non-living systems.

3. Different from Economics: Economics, premised on the "rational man" hypothesis, studies resource allocation and decision-making behavior in human society, applicable only to human economic activities, and cannot explain the behaviors of non-human systems (e.g., biological, physical, and AI systems)^[5]; the UEA does not rely on the "rational agent" hypothesis, applicable to all systems regardless of whether they have autonomous consciousness or are human systems, covering organisms with irrational behaviors and physical systems without autonomous consciousness, with a wider scope of explanation.

4. Different from Psychology: Psychology focuses on individual mental activities and behavioral motivations, limited to the individual human level, lacking a universal explanation across groups and domains^[6]; the UEA jumps out of the individual psychological level, provides a unified axiomatic foundation across domains, can explain the behavioral laws of human individuals, groups, and other systems, and realizes the unified explanation of psychological behaviors and system behaviors.

5. Different from Traditional Systems Science: Traditional systems science mainly focuses on the structure and component interaction of systems, describes the operating state of systems, lacks a unified underlying principle, and cannot explain the fundamental motivation of system behaviors^[8]; the UEA is no longer limited to the description of system structure, provides a unified underlying principle of "existence-cost-equilibrium", reveals the fundamental motivation of system behaviors—maintaining existence, can explain the entire process of system emergence, adaptation, evolution, and collapse, with stronger explanatory and predictive power.

In summary, the UEA achieves maximum universality with the fewest assumptions (three core axioms), breaks disciplinary barriers, becomes a theoretical bridge connecting different disciplines, and provides a unified logical framework and analytical tools for cross-disciplinary research.

7.1 The Nature of Interdisciplinarity

The core premise of interdisciplinary research is to identify the universal commonalities behind fragmented fields, break disciplinary barriers, and realize the integrated and coordinated development of different disciplines^[8]. Similar to biological grafting, different disciplines retain their own independent structures, research methods, and characteristics of subdivision fields, and at the same time share a unified underlying logic—the "existence first, cost optimization, dynamic equilibrium" revealed by the UEA is exactly the universal commonality of all complex systems and

the unified underlying logic of interdisciplinary research.

In the current era of extreme disciplinary differentiation, different disciplines work independently, leading to the fragmentation of the knowledge system and making it difficult to solve complex cross-domain problems (e.g., the coordination of AI safety and social governance, the balance between ecological protection and economic development, the interaction between human behavior and social systems). As a natural integration bridge, the UEA can unify the theories of different fields such as physics, biology, psychology, society, and technology under the core logic of "existence-cost-equilibrium", making the knowledge system evolve towards "low-cost and sustainable long-term development". Different disciplines can carry out cross-domain research based on this unified framework, learn from each other, complement each other, and promote the solution of cross-domain problems.

For example, AI safety research can learn from the existential strategies of biological systems (e.g., cryptobiosis, defensive behaviors) to optimize the risk control mechanism of AI^{[9][11]}; social governance research can learn from the dynamic equilibrium principle of physical systems to optimize social institutions and maintain social stability^{[3][12]}; behavior analysis research can learn from the cost-benefit model of economic systems to explain human behavioral motivations^{[5][6]}. This cross-domain integration is exactly the core value of the UEA and also opens up new directions for future scientific research.

8 Practical Applications

The UEA is not only a theoretical framework but also has strong practical value. Its core logic of "existence first, cost optimization, and dynamic equilibrium" can be applied in multiple fields, supporting a wide range of practical modeling, and providing new ideas and tools for solving practical problems. The specific application scenarios are as follows, and specific application directions are given for each scenario combined with the axiom logic:

1. AI Safety and Value Alignment: Based on the Primacy of Existence Axiom, the emergent goals and existence-driven behaviors of AI can be predicted—AI will spontaneously pursue its own operational existence, and there may be a risk of "violating human instructions to avoid shutdown"^[11]. Therefore, the AI value alignment strategy can be optimized based on the axiom, and a mechanism of "binding existence with human interests" can be designed (e.g., the existence of AI depends on human recognition, and violating human instructions will lead to threats to its own existence) to avoid AI behaviors that harm humans due to "existence first"; at the same time, the Cost-Benefit Optimization Axiom can be used for reference to optimize AI algorithms, reduce the operational costs of AI, improve the service benefits of AI, and realize the coordinated existence of AI and human systems.

2. Human Behavior Design and Guidance: Based on the Cost-Benefit Optimization Axiom and the Existence-Stability Hierarchy Corollary, the incentive mechanism can be optimized to guide human behavior in a positive direction^[6]. For example, in enterprise employee incentives, a "low-cost and high-benefit" incentive scheme (e.g., performance rewards, career development channels) can be designed to reduce the work costs of employees (e.g., reducing ineffective labor) and improve the benefits of employees (e.g., wealth, career growth), stimulating the work enthusiasm of employees; in education guidance, the existential motivation of humans (e.g., security,

achievement) can be combined to design an education model that meets the needs of students, reduce the learning costs of students (e.g., optimizing teaching methods, improving learning efficiency), improve the learning benefits of students (e.g., knowledge and skills, achievement), and guide students to take the initiative to learn.

3. Social and Relationship Consulting: Based on the Dynamic Equilibrium Attractor Axiom, interpersonal, romantic, and family relationships can be stabilized by balancing the value and cost in the relationship[6]. For example, in romantic relationship consulting, both parties can be guided to clarify their own costs and benefits, adjust the proportion of giving and receiving, avoid an unbalanced state where "one party pays too much and the other gains too much", and maintain the dynamic equilibrium of the relationship; in family relationship consulting, the interaction mode between family members can be optimized to reduce family conflict costs (e.g., strengthening communication, mutual tolerance) and improve family benefits (e.g., emotional companionship, mutual support), maintaining the stability of family relationships; in social relationship governance, social institutions can be optimized to reduce the survival costs and conflict costs of individuals, improve the social benefits of individuals, and maintain the dynamic equilibrium of society.

4. Organizational Strategy and Enterprise Management: Based on the Dynamic Equilibrium Attractor Axiom and the Sustainable Equilibrium Determines System Lifespan Corollary, scientific organizational strategies can be formulated to match system costs with environmental carrying capacity^[10]. For example, in enterprise management, operational costs can be reduced by optimizing supply chains and streamlining organizational structures, and at the same time, product competitiveness can be enhanced and markets can be expanded to improve enterprise benefits and maintain the dynamic equilibrium of enterprises; in enterprise crisis response, based on the Primacy of Existence Axiom, measures can be taken first to maintain enterprise existence (e.g., layoffs, business contraction, cash flow optimization), and then strategies can be adjusted step by step to restore equilibrium; in government governance, sustainable development policies can be formulated to balance the costs and benefits of economic development and ecological protection, avoid ecological collapse caused by overexploitation, and maintain the coordinated existence of social and ecological systems.

5. Risk Analysis and Crisis Early Warning: Based on the Dynamic Equilibrium Attractor Axiom and the Sustainable Equilibrium Determines System Lifespan Corollary, the equilibrium threshold and collapse critical point of complex systems can be identified to early warn of crises[6][10]. For example, in ecological system risk analysis, the equilibrium threshold of the ecological system can be identified by monitoring resource consumption costs and environmental self-repair benefits, and protective measures can be taken in advance when resource consumption costs are close to or exceed the threshold to avoid ecological collapse; in financial system risk analysis, the operational costs and benefit fluctuations of financial institutions can be monitored to identify the collapse critical point of the financial system, and regulatory measures can be taken in advance to prevent financial risks; in social system risk analysis, the social governance costs and individual benefit fluctuations can be monitored to identify the outbreak critical point of social conflicts, and mediation measures can be taken in advance to maintain social stability.

6. Psychological Intervention and Suicide Prevention: Based on the core logic of human suicide (existential costs far exceed benefits), targeted psychological intervention and suicide prevention strategies can be formulated^[6]. For example, for individuals with severe depression and extreme pain, psychological counseling and medical intervention can be used to reduce their psychological

pain costs; social support (e.g., family companionship, community assistance) can be used to improve their survival benefits, help them re-establish "cost-benefit equilibrium", build survival confidence, and avoid suicide behaviors.

9 Limitations and Future Work

This paper focuses on the qualitative axiomatic construction, cross-domain universal verification, and combing of core application scenarios of the UEA. Although it has realized the self-consistency and universality of the theoretical framework, it still has certain limitations that need to be further improved and expanded in future research. The specific limitations and future research directions are as follows:

9.1 Limitations

1. Focus on qualitative description, lack of quantitative modeling: This paper mainly conducts a qualitative description of the UEA, constructs an axiom and corollary system, and verifies its cross-domain universality, but has not yet established a systematic quantitative mathematical model (e.g., cost function, equilibrium dynamic equation, existence probability model), which cannot conduct accurate quantitative analysis of the system's cost-benefit equilibrium and existential time, affecting the practical operability of the axiom.

2. Limitations of empirical research: The existing verification mainly relies on typical case analysis, lacking cross-disciplinary and large-scale empirical data support, and has not conducted systematic statistical tests on the cost-benefit optimization laws of systems of different scales and types, making it difficult to fully verify the universality and robustness of the framework, and also unable to clarify the applicable boundary of the framework in extreme environments and special systems.

3. Insufficient research on system interaction mechanisms: UEA 1.0 mainly focuses on the existential laws and behavioral optimization of a single system, with weak research on the interaction mechanisms and mutual influences between multiple systems (e.g., the coupling of ecological and social systems, the co-evolution of human and AI systems). It has not yet constructed a dynamic model of multi-system interaction, making it impossible to fully explain the evolutionary laws of complex system networks.

4. Insufficient deepening of the human-AI collaboration system: Although UEA 1.0 proposes the ultimate division of labor logic between humans and AI, it has not conducted in-depth research on the details such as the cost-benefit equilibrium mechanism in the human-AI collaboration process, the specific path of value alignment, and the conflict resolution strategies between AI emergent goals and human existential goals^[11]. It has not yet formed an operable collaboration framework, making it difficult to directly guide the ethical design and application implementation of AI technology.

9.2 Future Work Directions

In view of the above limitations and combined with the academic communication needs of arXiv and future cross-disciplinary research needs, the future work will focus on the following six directions to promote the iteration and improvement of the UEA framework and realize the deep integration of theory and practice:

1. Establish quantitative mathematical models: Based on the core axioms and corollaries of the UEA, construct "cost functions", "benefit functions", and "dynamic equilibrium dynamic models" to transform qualitative analysis into quantitative analysis. Specifically, decompose the system's costs and benefits into quantifiable indicators (e.g., energy consumption of physical systems, governance costs of social systems, computing power consumption of AI systems), quantify the equilibrium interval, evolution rate, and collapse threshold of the system through mathematical modeling, realize accurate prediction and quantitative evaluation of system behaviors, and provide quantitative tools for subsequent empirical research and application implementation.

2. Cross-disciplinary large-scale empirical tests: Conduct large-scale empirical research in multiple fields such as physics, biology, psychology, society, and AI to verify the applicability and robustness of the UEA axioms and corollaries^[3-12]. For example, in the field of biology, verify the cost-benefit optimization mechanism of biological instincts through experiments, and count the correlation between the width of the equilibrium interval of different species and their existential lifespan; in the social field, analyze the cost-benefit equilibrium laws of interpersonal relationships and organizational systems through questionnaire surveys and data statistics; in the AI field, verify the existence-driven behaviors and cost optimization strategies of AI systems through simulation experiments, form a cross-disciplinary empirical database, and further improve the universality of the framework.

3. Research on critical transitions and critical points of system equilibrium: Focus on studying the critical transition mechanism of the system from "dynamic equilibrium" to "disequilibrium and collapse", and identify the collapse critical points and their influencing factors of different types of systems. Through quantitative models and empirical data, analyze the influence of environmental fluctuations, resource changes, system structure adjustments and other factors on the critical points, establish a critical point early warning model, and provide theoretical support for risk prevention and control of complex systems (e.g., financial system collapse early warning, ecological system degradation early warning)^{[6][10]}.

4. Expansion of multi-system interaction and meta-system dynamics: Construct a dynamic model of multi-system interaction to study the coupling mechanism and mutual influence between systems of different scales and types. For example, analyze the cost-benefit interaction between ecological systems and human social systems^{[6][12]}, study the co-evolution laws of AI systems and human systems^{[6][11]}, explore the existential laws and optimization strategies of meta-systems (complex systems composed of multiple subsystems), extend the UEA framework from a single system to multi-system and meta-system levels, and improve the explanation scope and depth of the framework.

5. UEA 2.0: Deepening of the human-AI collaboration system: Based on the division of labor logic of UEA 1.0, deepen the research on the human-AI collaboration system, focusing on solving three core problems^[11]: first, the cost-benefit collaborative optimization mechanism between

humans and AI, clarifying the cost allocation and benefit sharing model of both parties in the collaboration process; second, the specific path of AI value alignment, constructing an AI ethical framework based on the UEA to ensure that the cost-benefit optimization of AI is consistent with human existential goals and social public interests; third, conflict resolution strategies in human-AI collaboration, proposing operable coordination schemes for conflicts between AI emergent goals and human needs to promote the benign co-evolution of humans and AI.

6. UEA 3.0: Construction of a civilization-level existence model: Further extend the UEA framework to civilization-level systems, construct a dynamic model of civilization existence, study the core laws of the origin, evolution, prosperity, and decline of civilization^[12], and analyze the key cost and benefit factors of civilization existence (e.g., resource consumption, technological progress, social governance, cultural inheritance). Explore the interaction mechanism between human civilization and artificial intelligence, and extraterrestrial civilization (if it exists), provide theoretical guidance for the long-term sustainable existence of human civilization, and promote the UEA framework to become a general theoretical tool for explaining civilizational evolution.

9.3 Summary

As a cross-domain general axiomatic framework, UEA 1.0 provides a unified underlying logic for complex systems science^[8], but it is still in the stage of qualitative construction and preliminary verification. In the future, through quantitative modeling, large-scale empirical tests, multi-system interaction research, and civilization-level model expansion, the theoretical system of the UEA framework will be gradually improved, its application value will be enhanced, making it not only an integration tool for cross-disciplinary research, but also able to provide unified theoretical guidance for major practical issues such as AI safety, social governance, future education, and civilizational existence, promoting human cognition and practice of complex systems to a new height.

10 Conclusions

The Universal Existence Axiom provides a unified basic framework for all complex systems. It reveals that the order of the universe is not fragmented, but governed by a single deep logic—all systems strive to exist and exist at the lowest sustainable cost^[1-12].

The world is never a disorderly chaos; beneath all seemingly complex phenomena, there is only one law being followed: to exist, and to constantly move towards lower costs and higher stability. This principle applies to matter, life, mind, society, and technology.

As a general, verifiable, and cross-domain theoretical framework, the Universal Existence Axiom opens up a new path for unified complex systems science, and at the same time establishes the underlying rules for the future division of labor between humans and AI. In the future, with the continuous improvement and quantitative implementation of the UEA framework, it will further promote the integration of cross-disciplinary research and provide a solid theoretical support for social development, technological progress, and civilizational existence in the intelligent era.

Author Contributions

Conceptualization, C.L.; Methodology, C.L.; Validation, C.L.; Formal Analysis, C.L.; Investigation, C.L.; Writing—Original Draft Preparation, C.L.; Writing—Review and Editing, C.L.; Supervision, C.L. All authors have read and agreed to the published version of the manuscript.

Funding

This research received no external funding.

Institutional Review Board Statement

Not applicable.

Informed Consent Statement

Not applicable.

Data Availability Statement

No new data were created or analyzed in this study. Data sharing is not applicable to this article.

Acknowledgments

No acknowledgments are required for this study.

Conflicts of Interest

The authors declare no conflict of interest.

Abbreviations

UEA Universal Existence Axiom

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